NUMBER SYSTEM MADE EASY

0101 1010 0011

1100 0110 1001

VIDEO 1

Introduction to Number Systems

Grade 9 • Digital Technology and Innovation

Understand the systems behind the numbers computers use.

Today’s learning goals

By the end of this lesson, you should be able to...

IDENTIFY

The four main number systems

EXPLAIN

What base or radix means

UNDERSTAND

How positional values work

Let’s start with the basics.

What is a number system?

A way of representing numbers using a specific set of digits and rules.

EVERYDAY EXAMPLE

Decimal numbers such as 10, 25, 100 and 500.

COMPUTING CONNECTION

Computers represent and process information using digital number systems.

Different systems use different sets of symbols.

Base or radix

The base tells us how many unique digits or symbols are available.

BINARY

BASE 2

Digits: 0, 1

OCTAL

BASE 8

Digits: 0–7

DECIMAL

BASE 10

Digits: 0–9

HEXADECIMAL

BASE 16

Digits: 0–9, A–F

Base = number of unique digits / symbols used

1. Binary number system

Base 2 • Uses only two digits: 0 and 1

0 1

EXAMPLE

101101_2

The small 2 tells us the number is written in base 2.

Binary is important in computing because digital systems can represent information using two states.

2. Octal number system

Base 8 • Uses the digits 0 through 7

0 1 2 3 4 5 6 7

VALID OCTAL

725_8

NOT VALID OCTAL

728_8 • digit 8 is not allowed

There is no digit 8 or 9 in the octal system.

3. Decimal number system

Base 10 • The number system we use most often in everyday life

0 1 2 3 4 5 6 7 8 9

EXAMPLE

572_{10} • An ordinary number such as 572 is normally understood to be decimal.

4. Hexadecimal number system

Base 16 • Uses 0–9 and six letters A–F

0 1 2 3 4 5 6 7 8 9

A B C D E F

A	B	C	D	E	F
10	11	12	13	14	15

Example: $2F_{16}$

Compare the four systems

Notice how each system has a different base and set of digits.

NUMBER SYSTEM	BASE	DIGITS
BINARY	2	0–1
OCTAL	8	0–7
DECIMAL	10	0–9
HEXADECIMAL	16	0–9, A–F

Positional values

A digit’s value depends on the digit itself and its position.

572_{10}

5

Hundreds

$5 \times 10^2 = 500$

7

Tens

$7 \times 10^1 = 70$

2

Units

$2 \times 10^0 = 2$

500 + 70 + 2 = 572

The same idea applies to other bases: powers of 2, 8 or 16.

Why does this matter in computing?

Binary is the foundation; octal and hexadecimal make long binary values more compact.

BINARY

Used by computers to represent information.

OCTAL

A compact representation of binary values.

HEXADECIMAL

A compact representation of binary values.

Understanding number systems helps us understand how computers represent and process information.

Quick quiz

Pause and choose your answer before revealing it.

Which number system uses only 0 and 1?

A
Decimal

B
Octal

C
Binary

D
Hexadecimal

ANSWER: C — BINARY

One more check + lesson recap

Keep these key facts in mind.

QUESTION

Which number system has base 16?

Answer: HEXADECIMAL

TODAY WE LEARNED

Number system • Base / radix
Binary 2 • Octal 8
Decimal 10 • Hexadecimal 16
Positional values

Next: Binary ↔️ Octal — get ready to group those binary digits!

THANK YOU!